## Motivation

As engineers, we always try to model the manufacturing processes in a way that optimizes the output. We often have limited knowledge about the underlying process mechanisms (process know-how), the exact functional form of the response, and the design variables. Therefore, we perform experiments or simulate the process to gain insights. However, experiments are sometimes tedious and expensive. One way to alleviate this burden is to build an approximate model known as the **surrogate model**, e.g., Response Surface Method (RSM), meta-models, and emulators. A model is constructed based on modeling the simulator's response to a limited number of reasonably chosen data points. This approach is also known as black-box modeling.

## Aim

The objective of this competition is to **optimize one response variable** while simultaneously satisfying specifications on other response variables. The competition expects participants to leverage various mathematical and statistical models to analyze a limited number of experimental or simulated data points and achieve the desired optimum output.

## Competition Details

In this competition, you will work on the layout of a process plant, a decision that has a major impact on overall economics. A good plant layout must achieve cost minimization, maximize safety (minimize risk), and satisfy other practical constraints. Here, costs include land, piping, accident, and material-handling costs, while safety-related risk is considered at both the individual and societal levels.

We define a total cost function T that combines both cost and risk effects, but its exact functional form is unknown to you. The **function T** depends only on two independent variables: **C (a measure of cost)** and **R (a measure of risk)**. During the competition, we will provide you with **up to 20 data points**, i.e., values of T at selected combinations of C and R. Obtaining each value of T is assumed to be difficult and time-consuming, which mimics a realistic industrial scenario.

Your challenge is to use these **limited data points** to construct a suitable surrogate/optimization strategy and **minimize T** over the ranges **C [275,331]** and **R [59, 76]** (with T, C, and R all expressed in the same units). Your final objective is to report the minimum value of T you obtain, along with the corresponding optimal values of C and R.

# Structure of Competition

**Optimization Problem**

Participants will be provided with the Optimiser **Simulator App**, which computes the value of the total cost T. Teams can input values of C and R and obtain the corresponding value of T as output. The simulator link and setup instructions will be shared with all participants via email and the official WhatsApp group within one week of the competition launch (expected by 27th February).

Each team must use the simulator to strategically query the system (maximum query limit will be specified in the problem statement) and determine the optimal values of C and R that minimize T.

**Submission Requirements**

Each team must submit ONE combined submission consisting of:

1. **Final Report (PDF)**
2. **Code / Notebook File**

You may use any of the following tools: Python, MATLAB, R, Excel, or any equivalent software.

Only one team member should submit the final solution.

Multiple submissions are not allowed and will lead to disqualification.

**Report Guidelines**

The report must document the complete optimization workflow and include:

- **Team Details**: Team ID; names and contact details of all members.
- **Experimental Data**: Table of all queried data points with corresponding values of C, R, and T.
- **Methodology**: Data selection strategy, exploration/refinement approach, and handling of noise.
- **Modelling & Analysis**: Surrogate model used, mathematical formulation, relevant equations, plots, tables, calculations, and description of the optimization method.
- **Final Results**: Predicted optimal values of C and R, and the corresponding minimum value of T.

**Formatting Requirements**: The report must be neat and well-structured, with numbered figures and tables (each with captions), and clear written explanations supporting the results.

## Evaluation Criteria

The final score will be based entirely on the submitted report, code, and simulator query logs.

**A. Data Strategy — 30 marks**

Participants are expected to choose simulator data points intelligently. Each simulator run represents an expensive experiment; therefore, teams should aim to extract maximum information with minimal runs. Evaluation will focus on how effectively teams explore the domain and refine their search.

**B. Technical Analysis — 45 marks**

Participants must apply appropriate mathematical and statistical tools to build a surrogate model and analyze the dataset. Marks will be awarded for the quality of modelling, correctness of methodology, and effectiveness of the optimization approach.

**C. Accuracy of Final Solution — 20 marks**

Marks will be awarded based on how close the submitted optimum is to the true global minimum:

- Within 1% of true minimum: 20 marks
- Within 3%: 15 marks
- Within 5%: 10 marks
- Within 10%: 5 marks

**D. Report Quality — 5 marks**

Clarity, organization, proper plots/tables, mathematical rigor, and overall presentation quality.

**Query Efficiency Bonus**:

Participants are allowed a maximum of 20 simulator runs. To reward efficient use of the query budget, teams will receive a bonus of $(20 - X) \times 1$ marks, where X is the number of simulator runs used. The maximum **bonus is 10 marks** (achieved by using 10 or fewer runs). This bonus is included within the Data Strategy category.

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# OPTIMISER

## Certificates and Prizes

1. The **top five** teams will receive cash prizes totaling **INR 15k**.
2. The **top 10** teams will receive a **Certificate of Appreciation**, and a Certificate of Participation will be awarded to **all teams** making a valid final submission.

## Teams Specifications:

1. A maximum of 2 participants are allowed per team.
2. Open to undergraduate and postgraduate students.
3. Students of different institutions can also form a team.
4. At least one team member must be from a Chemical engineering background.
5. Participants are permitted to register for only one team. Any participant found registered with multiple teams will result in the immediate disqualification of all teams involved.

Step 1: Fill the registration form by **27th February**.
Step 2: **Submit** the final submission form by **4th March** with your Team name and Report including all your findings.

**Note:**
AZeotropy, IIT Bombay, reserves the right to modify any rules. Updates will be communicated via the official WhatsApp group and email.

For any queries, please feel free to contact:
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